

Qatar-4b

R-2627

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_191208 · created 2026-08-29T00:00:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458825.7065 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.052
Transit depth (Rp/Rs)^2	2.24 +/- 1.56 [%]
Orbital Inclination (inc)	89.88 +/- 2.15
Airmass coefficient 1 (a1)	0.58 +/- 0.26
Airmass coefficient 2 (a2)	0.46 +/- 0.73
Scatter in the residuals of the lightcurve fit is	42.44 %
Best Comparison Star	#2 - [21, 391]
Optimal Aperture	3.0
Optimal Annulus	12.94
Transit Duration (day)	0.0743 +/- 0.0039

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.15 $\pm$ 0.052 vs 0.138 (deviation 0.17 σ)
Duration fit vs published	0.0743 d vs 0.0892 d (deviation 2.84 σ)
Mid-time O-C	2.7 min
Depth SNR	2.1
Residual scatter	42.44 %

Light curve

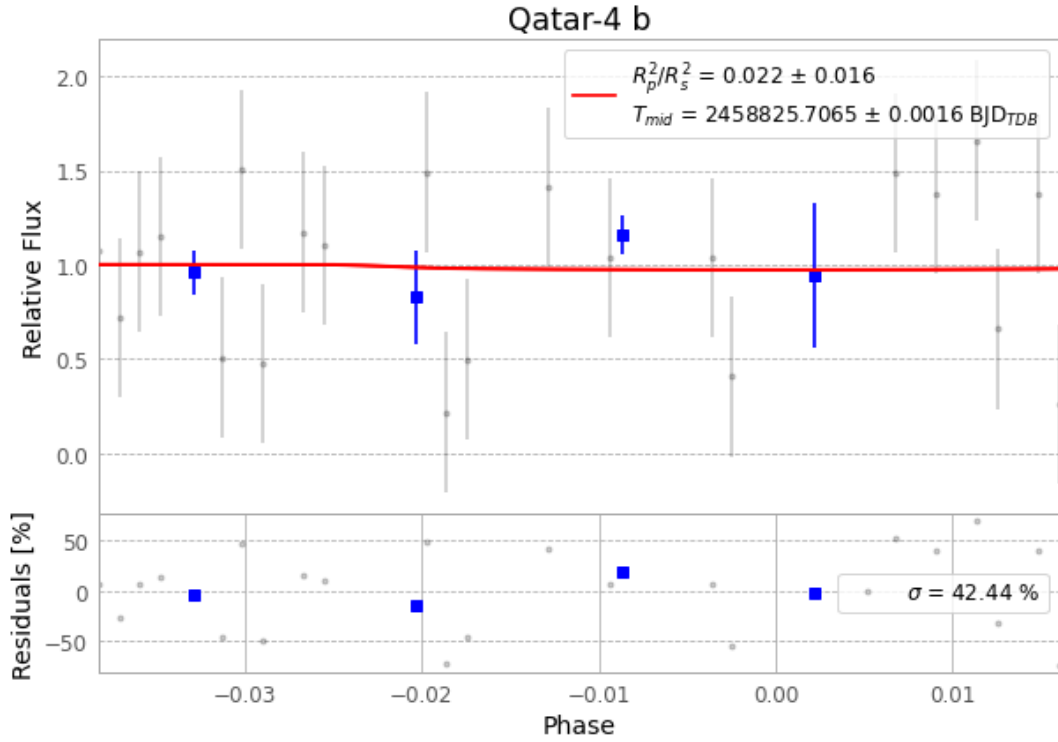


Figure 1 — FinalLightCurve_Qatar-4 b_2019-12-07.png (SHA-256 703a23697b5ccc3b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [233, 104], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 50d9853f190cb4df...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), Qatar-4191208030612.FITS → Qatar-4191208053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-191208.log (d6250dda291f...), FinalLightCurve_Qatar-4 b_2019-12-07.png (703a23697b5c...), FinalLightCurve_Qatar-4 b_2019-12-07.csv (aa0b88d53354...)

Compute resources

Wall time 133.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 00:00Z.